

Performance: Synthetic Tests

HD Tach RW

A drive consuming higher CPU resources is not a good thing, since it tends to slow down the computer. A CPU utilisation of above 15 per cent generally indicates something seriously wrong. However, if the figure hovers around 3 to 5 per cent, it shouldn't really make a difference with today's CPUs. This is especially true for external hard disks—the disk isn't connected to the computer all the time, and one doesn't usually do other tasks while transferring data to or from the drive.

In the HD Tach RW test, the CPU utilisation scores of the Freecom FHD-XS, the LaCie Skwarim, and the LaCie Rugged were all low—3 to 4 per cent. The Seagate 40 GB, however, went up to 9 per cent.

A lower random access time means better all-round performance. The Seagate 40 GB drive had the lowest random access time of 17.5 ms, closely followed by the Transcend StoreJet 80 GB with 17.8 and the LaCie Rugged with 17.9 ms.

In the average read and write speed tests, the LaCie Rugged, which posted the highest average read score of 30.3 MBps, was also the one to score the lowest in the average write test

with just 3.9 MBps. The WD Passport scored the highest in the read test with 25.5 MBps; closely following it were the Freecom FHD-2 PRO and LaCie Design by F. A. Porsche, each with 25.3 MBps.

SiSoft Sandra 2007 Engineer

After testing the raw hard drives, we formatted them with the NTFS file system to evaluate their file system performance. (SiSoft Sandra benchmarks drives after they are formatted using a file system.)

The LaCie Design by F. A. Porsche and the WD Passport were tied for the top spot with a drive index of 27. Next-best was the Freecom FHD-2 PRO with 25. These drives performed extremely well in all the tests under this benchmark. The LaCie Rugged behaved strangely in this test as well: while it posted average scores in the read tests, it performed below our expectations in the write tests.

The access times of most of the drives ranged between 10 and 15 ms. The exception was the Seagate 40 GB drive, which scored 19 ms. Of course, a lower access time is desirable for faster file access.

Performance: Real-World Tests

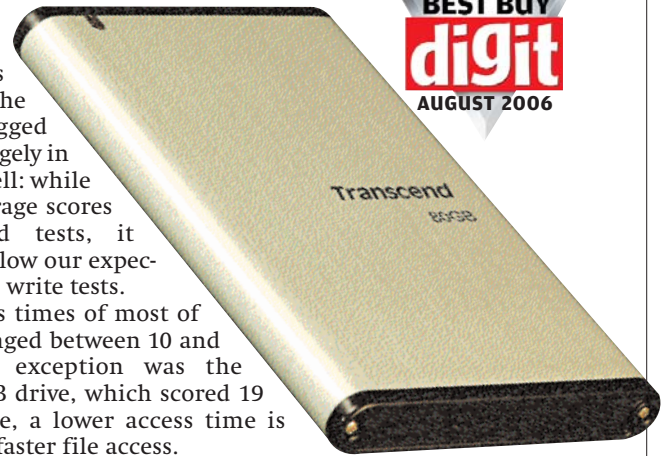
Most of the data we usually deal with comprises assorted files. While transferring 1 GB of assorted files from our primary drive to the external drive (this is the real-world write performance test for the drive), the Transcend StoreJet 80 GB logged the best time: 55 seconds. The LaCie Rugged came in last at 110 seconds—much slower than any of the drives in this category.

The picture was different when the assorted files were transferred from the external drive to our primary hard drive (this is the real-world read performance). This time, the WD performed well, requiring just 43 seconds for the job, while the LaCie Design by F. A. Porsche was next with 44 seconds. The Freecom FHD-XS was the slowest here with 83 seconds.

When we wrote a 1 GB sequential file to the drives, the LaCie Design by F. A. Porsche was the winner, clocking just 40 seconds. Yet again, the LaCie Rugged was the slowest in the real-world write performance, taking 173 seconds.

In the test where we read a 1 GB sequential file from the external drive to our primary drive, the LaCie Rugged and the LaCie Design by F. A. Porsche were the fastest, each taking 33 seconds. The slowest drive was again the Freecom FHD-XS at 94 seconds.

When you need to transfer large archives, movies, CD or DVD images and so forth, you are transferring sequential data. The above test should give you an idea of which drive is better suited for such operations.



Transcend
StoreJet 2.5 80 GB

Read, Sequential Write, Random Write and Access Time scores were reported at the end of this benchmark, which we noted.

Real World Test: File Copy

The real-world tests consisted of copying a 1 GB file (for sequential data transfer speed) and 1 GB of assorted files (for random data transfer speed) from one partition of the hard drive to the other. The time taken for this data transfer was noted. The sequential file transfer time will give a fair idea about how the hard drive will perform when large files such as movies are transferred. The assorted files consisted of multiple file types that included applications, Word documents, Excel files, MP3s, small video clips, etc., thus simulating a real-world situation.

The tests we carried out were:

- ☐ Time taken to copy the file(s) to the external drive, thus giving a measure of the drive's interface performance as well as the write performance.
- ☐ Time taken to copy the file(s) from the external drive, thus giving a measure of the drive's interface performance as well as the read performance.
- ☐ Time taken to copy the file(s) within the external drive, thus giving a measure of how well-tuned the drive's internal mechanics are, well as the overall performance.

Real World Test - Multitasking

In this test, we copied 1 GB of assorted data from the external hard drive and to it simultaneously. This tests the drive's multitasking capability and also gives a measure of its interface performance.